Be the Change You Want to See: Revisiting Remote Sensing Change Detection Practices

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***Abstract*— Remote sensing change detection aims to localize semantic changes between images of the same location captured at different times. In the past few years, newer methods have attributed enhanced performance to the additions of new and complex components to existing architectures. Most fail to measure the performance contribution of fundamental design choices, such as backbone selection, pretraining strategies, and training configurations. We claim that such fundamental design choices often improve performance even more significantly than the addition of new architectural components. Due to that, we systematically revisit the design space of change detection models and analyse the full potential of a well-optimized baseline. We identify a set of fundamental design choices that benefit both new and existing architectures. Leveraging this insight, we demonstrate that when carefully designed, even an architecturally simple model can match or surpass state-of-the-art performance on six challenging change detection datasets. Our best practices generalize beyond our architecture and also offer performance improvements when applied to related methods, indicating that the space of fundamental design choices has been underexplored. Our guidelines and architecture provide a strong foundation for future methods, emphasizing that optimizing core components is just as important as architectural novelty in advancing change detection performance. Code is at: https://github.com/blaz-r/BTC-change-detection**

***Index Terms*— Change detection, method optimization, remote sensing, supervised learning.**

I. Introduction

REMOTE sensing change detection aims to detect and localize semantic changes between images of the same region. It is one of the fundamental tasks in remote sensing, providing insights into natural and human-driven processes such as deforestation, urban expansion, and natural disasters [1], [2], which have a significant impact on climate, ecosystems, and human life [3], [4], [5]. As remote sensing data availability grows, so does the need for efficient and accurate change detection methods.

Change detection can be approached in various ways, but the dominant paradigm in remote sensing focuses on the bitemporal setting in which a pair of images is analyzed. Recently proposed deep-learning-based change detection methods [1], [6], [7], [8], [9], [10] largely attribute performance gains to complex architectural innovations and fail to measure the performance contributions of fundamental design choices like backbone selection and training configurations. We hypothesize that a significant part of the observed performance improvements comes from making the correct fundamental design choices rather than architectural innovations. On top of that, many methods nowadays still underutilise pretraining, employ inconsistent data augmentations, and rarely experiment with alternative training strategies despite recent advancements [11], [12], [13], meaning that there is an excellent opportunity for improvement in picking the right design choices. While some studies have also explored new backbone and training choices [1], [14], [15], [16], they often do so accompanied by larger architectural changes, making it difficult to quantify their standalone contributions. We hypothesize that carefully selecting and optimizing these elements significantly impacts change detection performance.

In our work, we build a model from scratch (starting from a simple baseline), varying only the base design choices. We iteratively refine the simple base model, independently examining each design choice's impact on the overall performance. More specifically, we focus on backbone architecture, backbone size, pretraining, data augmentation techniques, loss functions, and learning rate schedulers. While many of these techniques were studied in other domains [17], [18], [19], [20], such as pretraining on natural images,[1] they have not yet been comprehensively evaluated in change detection. This process leads to an optimized yet architecturally simple change detection model, which we refer to as Be The Change (BTC). After applying all gathered insights, the performance of BTC compared to the baseline model across six datasets increases by 9.4 percentage points (p.p.), as illustrated in Fig. 1. Furthermore, we show that applying our guidelines to existing remote sensing foundation models and change detection-specific architectures improves their performance.

We conduct extensive experiments on six standard change detection datasets to validate our insights and demonstrate their broad applicability. Our results show that a simple

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[1]The term *natural images* typically refers to images captured at ground-level and containing diverse real-world scenes, such as those found in the ImageNet dataset. We use this term to distinguish these images from the top–down view of satellite and aerial images.

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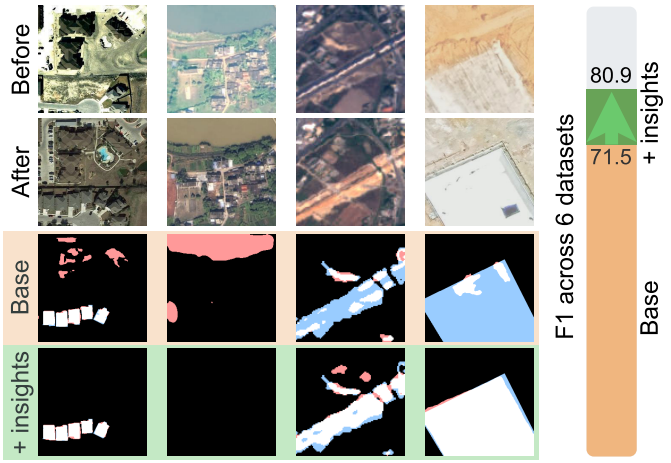


Fig. 1. Comparison of change detection performance. The first two rows contain the input image pair, the third row depicts predictions of the base model, and the last row depicts predictions of the optimized model, incorporating guidelines from our analysis. False positives (red) and false negatives (blue) highlight differences in prediction quality. The $F1$ score is reported across six different datasets. The improvement of 9.4% demonstrates how these fundamental yet often overlooked design choices greatly enhance performance. (More information are given in Section VI.)

but well-optimized BTC architecture achieves performance comparable to, or even surpassing, state-of-the-art methods, particularly on smaller datasets. This reinforces our claim that optimal fundamental design choices have been underutilized and that recent performance gains cannot be solely attributed to architectural novelty. The lack of awareness about the impact of these core components, coupled with the absence of systematic analyses, has likely contributed to this oversight. The contributions of this work are as follows.

1) We *systematically analyze* key design choices of base components in change detection methods. This includes various backbone aspects, pretraining, data augmentations, loss functions, and learning rate schedulers. Our findings highlight the overlooked impact of these fundamental components and help us *establish best practices* for their use in the future.
2) Our study goes beyond standard optimization by showing *generalization to related methods*. By applying our insights to remote sensing foundation models and other change detection methods, we show consistent improvements in their performance. This verifies that recent methods have underutilized these fundamental design principles and that architectural novelty alone does not account for observed improvements.
3) We propose BTC, an architecturally simple yet *powerful change detection model*. It leverages our observations to match or even outperform the current state-of-the-art. Due to its architectural simplicity, we believe it presents an excellent foundation for new architectures to build upon.

II. Related Works

Change detection has a long history, ranging as far as 50 years into the past [21]. Some of the early approaches used simple pixelwise comparisons, statistical methods, or traditional machine learning [22], [23]. The field has witnessed an increase in popularity with the introduction of deep learning methods. One of the first successful deep-learning-based approaches that outperformed traditional methods used convolutional architectures [2]. FC-Siam-Diff [24] introduced a Siamese network as the encoder, with the idea that the latent representations of the images in a pair will be aligned if there are no semantic changes. This architecture, consisting of two encoders with shared weights, remains a widely used foundation in modern change detection.

With the introduction of new architectural elements [25], [26], contemporary change detection methods have advanced by integrating transformer models [1], [8], [15] and other building blocks [7], [16], [27]. Many, however, still rely on CNNs for feature extraction and use various attention approaches to improve performance [9], [10], [28]. Given the challenge of fully training backbones from scratch on limited data [29], the majority of such methods rely on pretrained backbones, typically trained on ImageNet [30]. BAN [31] uses a remote sensing foundation model instead of a regular backbone to enhance the features used in its change detection architecture. However, the impact of backbone architecture and its pretraining is rarely evaluated [15], despite its significance that we show in this article.

Pretraining plays an important role in many remote sensing tasks [32], [33], with most approaches leveraging self-supervised contrastive learning (CL) [34], [35], [36] or masked image modeling (MIM) architectures [37], [38]. The authors of GFM [39] additionally explored the combination of the MIM approach with the simultaneous distillation of features from an ImageNet pretrained model. However, these strategies do not always translate to optimal change detection performance [40], [41]. While multitask pretraining (MTP) [40] has demonstrated strong results, its complexity and high computational cost can limit its practicality. Moreover, most pretraining approaches focus solely on representation learning rather than optimizing the training process specifically for change detection.

Addressing this, some methods have introduced pretraining tailored to change detection [36], [42], [44], [45]. Authors of the Changen method [45] proposed the generation of synthetic changes to create a large-scale dataset for pretraining, achieving good results. Despite these efforts, many specialized pretraining techniques struggle to consistently outperform general-purpose backbones pretrained on tasks like ImageNet classification [41].

Systematic analysis and optimization of training strategies was explored for convolutional networks [13], [46], large language models (LLMs) [47], and, to a limited extent, remote sensing pretraining [41], [48]. Corley et al. [49] addressed some of the issues with current change detection methodologies and their evaluation protocols, though without an in-depth analysis of underlying factors. Meanwhile, systematic research on architectural and training aspects for change detection remains sparse [29], [41]. Thia article thus contributes an extensive analysis with valuable insights into these challenges.

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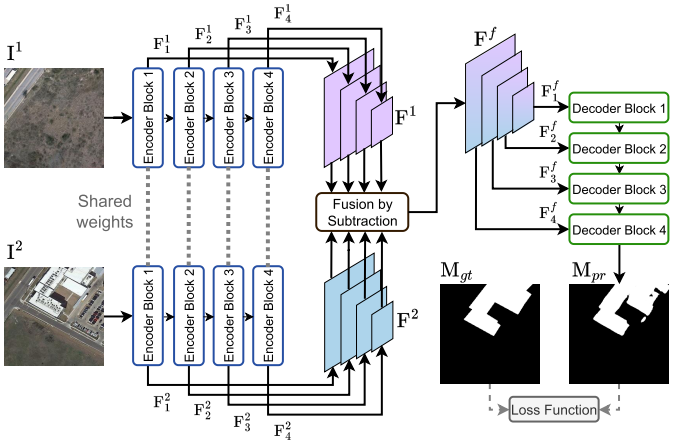


Fig. 2. Outline of our change detection method. Each image in the given pair (I^1, I^2) is processed using the same encoder. Then, the extracted features (F^1, F^2) are fused (in our case, by simple subtraction), and the resulting fused features F^f are decoded using the UPerNet decoder [51] to obtain the predicted change mask $\mathbf{M}_{\text{pr}}$.

III. Methodology

To clarify our evaluation approach, we first introduce the change detection framework we use in Section III-A. Next, we outline the evaluation protocol in Section III-B, covering datasets and metrics. Finally, we provide key implementation details in Section III-C.

A. Change Detection Framework

We use a simple architecture (as illustrated in Fig. 2) to evaluate the impact of different design choices. The model consists of three main components: 1) the *Siamese encoder* first extracts features from the prechange and post-change images; 2) a *feature fusion module* then combines the information from these features into a single representation; and 3) the *decoder* finally predicts the binary pixel-level change map based on the given representations.

A common choice for the *encoder* (also called *backbone*) are networks such as ResNet [50], ViT [25], and ConvNext [13]. Their key advantage is that they have pretrained weights available from various pretraining tasks (e.g. segmentation and classification). We primarily use Swin [11], a choice experimentally validated later in this article. The Siamese setup applies the same encoder with shared weights to both images, producing corresponding feature maps.

Rather than introducing additional fusion or decoder modifications, we focus on optimizing the encoder—a crucial yet often overlooked aspect of change detection models. While many methods utilize standard backbones and refine fusion and decoder modules [1], [6], [8], [10], [15], our analysis demonstrates that backbone choice alone leads to substantial gains, making it a critical design consideration for future models. We preliminarily evaluated some noncomplex fusion and decoder designs from prior work [24], [39], [40], and adopted well-established and best-performing choices to ensure robust and fair comparison.

The *feature fusion module* is implemented as a simple element-wise *subtraction* of features at the same hierarchical level: $F_i^f = F_i^1 - F_i^2$, where F^1 and F^2 are features from pre- and post-change images respectively, and i represents different layer levels from the encoder.

For the *decoder*, we use UPerNet [51], which efficiently processes multi-level fused features F^f to generate a change segmentation map. The change map is then thresholded at 0.5 to produce the final binary change mask $\mathbf{M}_{\text{pr}}$.

B. Evaluation Regime

Each experimental setup is evaluated on six different change detection datasets (as described in Section III-B1). For consistency, all experiments use the same set of hyperparameters (as provided in Section III-C), and all model parameters are optimized (i.e., the backbone is never frozen).

To ensure a fair comparison, each experiment is repeated with three different random seeds, and the average $F1$ across these runs is reported. We also include *standard deviation* in Supplementary Material S4. The evaluation is performed on the predefined testing set of each dataset using the model from the final epoch.

1) Datasets: For a comprehensive and robust evaluation, we select six commonly used RGB change detection datasets from various sensors, covering diverse change types, imaging conditions, ground sample distances (GSD), and dataset sizes. SYSU [52] is used as a candidate for changes of multiple types. Two datasets contain only building changes, the larger LEVIR [28] and the smaller EGYBCD [53]. GVLM [54] is used to evaluate landslide changes and CLCD [55] is a smaller dataset focusing on cropland changes. Finally, OSCD [2] is used to verify small-scale datasets with low-resolution Sentinel-2 imagery.

1) SYSU consists of 20 000[2] pairs of image patches with a high resolution of 0.5 m. It captures multiple change types, including buildings, groundwork, vegetation, and sea changes.
2) LEVIR contains over 10 000[2] image pair patches, captured over 5–14 years in U.S. $20 regions, with a resolution of 0.5 m. It is specifically designed for building change detection and includes only such labels.
3) EGYBCD is a smaller building change detection dataset with a little over 6000[2] image pairs. It features images with a high resolution of 0.25 m captured in Egypt between 2015 and 2022.
4) GVLM is tailored for landslide change detection. It contains over 7500[2] image pairs at a resolution of 0.59 m collected from 17 different regions worldwide.
5) CLCD targets cropland change detection. It includes approximately 2400[2] images collected between 2017 and 2019. It has varying resolutions from 0.5 to 2 m and focuses on visually challenging cropland changes.
6) OSCD is a low-resolution dataset with 10-m resolution images captured worldwide between 2015 and 2018 using the Sentinel-2 satellite. It consists of around

[2]Number of 256 × 256 patches.

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1200[3] image patch pairs, focusing only on urban changes.

More extensive details on these datasets can be found in Supplementary Material S1.

2) Metrics: Following standard practice [1], [8], [24], [40], [45], we evaluate change detection performance using the *binary F1* score, considering only *change class*. Some recent works deviate from this practice by treating the background as a separate class and computing the mean $F1$ score (mF1). Due to the inherent class imbalance, this inflates the $F1$ score and prevents fair comparison with other works.

For illustration, our best setup achieves an $F1$ of 80.9 % in the binary case, while mF1 reaches 89.6%. This discrepancy highlights why we urge researchers to adhere to the established protocol when implementing metrics.

A more in-depth discussion of the implementation and evaluation is present in Supplementary Material S2.

3) Proposed Evaluation Protocol: To promote consistency and reproducibility, we compile the above-stated details and outline the following recommended evaluation protocol for future change detection studies.

1) Use a diverse set of datasets covering different change types, resolutions, and acquisition settings.
2) Ensure consistent training hyperparameters across experiments.
3) Repeat each experiment with multiple random seeds and report the average performance.
4) Evaluate models on a fixed test set.
5) Use binary $F1$ score (focusing on the change class) to avoid performance inflation from class imbalance.
6) Fully disclose the design decision, even the fundamental choices.

Future methods can utilize our framework code (link is in the abstract), which already covers all the above-listed aspects.

C. Baseline Training Details

For our baseline, we start with Swin-T [11] as the encoder (*backbone*). Consistent with related work [1], [14], [15], [16], [28], the backbone is initialized with *ImageNet1k* [30] pretrained weights, unless stated otherwise. All other model weights follow the standard random initialization. Input images are cropped to 256 × 256 pixels, except for OSCD, where, following the original protocol [2], [24], images are cropped to 96×96 pixels and then rescaled to 256×256 pixels. The model is trained for 100 epochs with a batch size of 32 using the AdamW [56] optimiser with a base learning rate of 10^{-4} and with a weight decay of 10^{-4}. The only exception is OSCD, where we only train for 50 epochs due to a small training set. Following the literature [1], [8], [40], [45], the model is trained in a supervised fashion using binary cross-entropy (CE) loss, *with no scheduler* and *no augmentations*, unless stated otherwise. These hyperparameters remain constant throughout all experiments to ensure robust comparison.

[3]Number of 96 × 96 patches.

TABLE I
COMPARISON OF DIFFERENT PRETRAINING DATASETS FOR THE SWIN BACKBONE. NONE MEANS THAT NO PRETRAINING DATASET WAS USED. PAN STANDS FOR PANOPTIC AND SEM FOR SEMANTIC SEGMENTATION

	SYSU	LEVIR	EGYBCD	GVLM	CLCD	OSCD	*Avg*
None	77.0	88.2	62.7	76.7	37.0	87.5	71.5
ADE20k	81.5	**91.0**	**83.5**	**88.9**	74.5	50.9	78.4
City-Sem	**82.1**	**91.0**	83.1	**88.9**	**75.3**	**52.4**	**78.8**
COCO-Pan	81.7	90.8	83.3	88.5	74.6	49.8	78.1
IN1k	82.0	**91.0**	**83.5**	88.8	74.6	47.6	77.9
EuroSat	81.5	**91.0**	83.4	87.8	74.3	49.4	77.9

IV. BACKBONE DESIGN CHOICE ANALYSIS

The first set of design choices concerns the backbone, a cornerstone of many change detection methods that is often underutilized. We analyse the choice of pretraining, architecture, and model size in Sections IV-A–IV-C.

A. Pretraining Dataset

ImageNet1k [30] pretrained encoders are very common in related work [1], [14], [15], [16], [28], but whether they are the best-suited for change detection remains unclear. Swin offers multiple pretraining options, allowing us to assess the impact of dataset and task selection. We compare ADE20k [57] and Cityscapes [58] (semantic segmentation), COCO [59] (panoptic segmentation), ImageNet1k [30]—IN1k—(classification) and EuroSat [60], [61] (RS classification). Only the best-performing task-dataset pairs are reported, while complete results and implementation details are in Supplementary Material S3-A. Remote sensing foundation models are analyzed separately in Section VI-A.

Table I shows that segmentation pretraining outperforms classification, with Cityscapes-based semantic segmentation yielding the best results. Interestingly, EuroSat pretraining does not surpass ImageNet1k pretraining. Semantic segmentation pretraining likely enhances performance because change detection is a segmentation task. These findings suggest that pretraining for a similar task is more important than pretraining on domain data, which is an important insight that could guide future remote sensing pretraining schemes.

B. Backbone Architecture

ResNet remains a widely used backbone [1], [35], [36], but newer architectures offer potential improvements. We compare other common options: Swin [11], SwinV2 [12], ViT [25], ResNet18 [50], ResNet50 [50], and ConvNext [13]. All architectures were pretrained on ImageNet1k for the classification task (more details in Supplementary Material S3-C). The results in Table II reveal a significant influence of the choice of backbone architecture on overall performance.

We hypothesize that Swin outperforms others due to its small patch size and hierarchical design. This preserves high-resolution features but still enables efficient processing of global context [11].

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TABLE II
COMPARISON OF DIFFERENT BACKBONE ARCHITECTURES

	SYSU	LEVIR	EGYBCD	GVLM	CLCD	OSCD	*Avg*
Swin T	82.0	**91.0**	83.5	88.8	74.6	**47.6**	**77.9**
SwinV2 T	81.4	**91.0**	**84.1**	89.2	**75.0**	44.5	77.5
ViT T	**82.5**	84.5	81.3	88.0	73.5	40.3	75.0
ViT B	81.9	85.2	82.1	87.7	72.8	42.5	75.4
ResNet18	80.1	89.9	82.3	88.9	69.9	40.9	75.3
ResNet50	80.8	90.3	82.7	**89.5**	72.5	37.8	75.6
ConvNext B	81.5	90.7	83.7	88.8	73.0	40.2	76.3

TABLE III
COMPARISON OF DIFFERENT SWIN BACKBONE SIZES: TINY, SMALL, AND BASE

	SYSU	LEVIR	EGYBCD	GVLM	CLCD	OSCD	*Avg*
Swin T	**82.0**	91.0	83.5	88.8	74.6	**47.6**	77.9
Swin S	81.8	91.1	84.1	89.2	75.8	45.0	77.8
Swin B	**82.0**	**91.2**	**84.4**	**89.4**	**76.3**	47.3	**78.4**

C. Model Size

Larger backbones generally improve results in vision tasks given sufficient data and training [11], [25]. We validate this claim for the case of change detection by utilizing different Swin backbone sizes (specifics in Supplementary Material S2-C). Results in Table III show that the larger Swin-B achieves the highest performance, surpassing Swin-T by 0.8 p. p. on average. This improvement comes with almost double computational and memory costs, which could be acceptable in some critical applications.

V. TRAINING DESIGN CHOICE ANALYSIS

After investigating the backbone design choices, we now analyse common training techniques, including augmentations, learning rate schedulers, and different loss functions.

A. Augmentations

Larger backbones generally improve results and, in cases such as ImageNet classification, are almost standardized [11]. However, change detection lacks such consistency, so augmentation choices vary significantly. To assess their impact, we evaluate four common types: *Flip*, *Crop*, *Color jitter* (named Color in Table IV), and *Blur*, along with some promising combinations. Fig. 3 illustrates these augmentations. All augmentations have a 30% chance of being applied and are consistent for both images in a pair. Detailed augmentation parameters are provided in Supplementary Material S3-D.

The results in Table IV show that *Flip* and *Crop* improve performance (1.7 p. p. and 1.4 p. p. respectively). Their combination (*Flip* and *Crop*) also improves the baseline performance, but is less effective than only the *Flip* augmentation, likely due to the excessive distortion. An exception when using *Flip* occurs with the SYSU dataset, where dataset patches are already rotated beforehand [52], reducing the benefits of this augmentation.

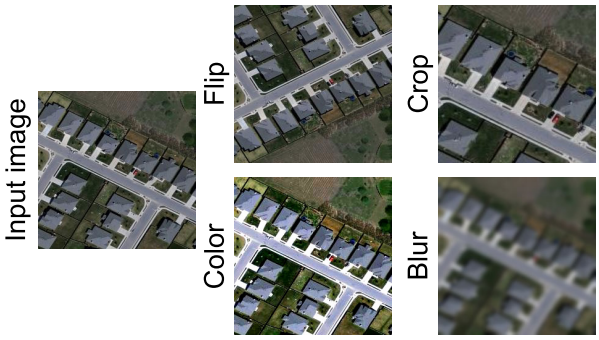


Fig. 3. Illustration of different augmentations used during the analysis. The leftmost image is the unaltered input image, while all others represent augmentations of the input image.

TABLE IV
COMPARISON OF DIFFERENT AUGMENTATION TECHNIQUES USED DURING TRAINING. *Flip* AND *Crop* SERVE AS DATA EXPANDERS, WHILE *Color* AND *Blur* CAN BE USED AS COUNTERMEASURES FOR PHOTOGRAPHIC VARIATIONS IN DATA

	SYSU	LEVIR	EGYBCD	GVLM	CLCD	OSCD	*Avg*
None	**82.0**	91.0	83.5	88.8	74.6	47.6	77.9
Flip	81.3	**91.4**	**85.4**	**90.0**	78.7	**51.0**	**79.6**
Crop	81.8	91.1	85.2	89.3	77.4	50.9	79.3
Flip & Crop	80.3	**91.4**	84.8	**90.0**	**79.3**	49.1	79.2
Color	81.7	91.0	83.7	89.0	74.4	47.9	77.9
Blur	81.4	91.1	83.6	88.9	74.8	46.5	77.7

Although *Color jitter* and *Blur* augmentations are sometimes used in related methods [1], [40], [45], our findings suggest that they do not widely help with performance.

We hypothesize that *Flip* and *Crop* play a similar role during training—extending the dataset. Since remote sensing data lacks a fixed orientation [62], *flipping* is a natural choice. The *Crop* augmentation also plays an important role, forcing the model to adapt to variations in GSD [38]. In contrast, the *Color jitter* and *Blur* augmentations appear to be more dataset-dependent, potentially serving as a countermeasure for photographic variations in images (e.g., brightness, contrast, and blur).

B. Learning Rate Scheduler

Learning rate schedulers are widely used to stabilize model training and improve performance, but no single choice dominates in change detection. We thus evaluate five common types: *multistep*, *cosine*, *exponential*, *linear*, and *polynomial*. Warm-up variants were also tested but did not yield improvements, so they are omitted. Implementation details are in Supplementary Material S3-E.

As shown in Table V, training without a scheduler achieves the best result. However, as we will later show, the *cosine* [63] scheduler proves beneficial when combined with augmentations. Our intuition is that it serves as a countermeasure toward the end of training for increased variability introduced by augmentations.

C. Loss

CE loss is the most widely used loss function in related works, but alternative choices, such as Dice loss [64] and Focal

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TABLE V

COMPARISON OF DIFFERENT LEARNING RATE SCHEDULERS. THE BEST-PERFORMING *Cosine* SCHEDULER PERFORMS WORSE THAN THE BASELINE, BUT AS WE SHOW IN TABLE VII, IT HELPS WHEN COMBINED WITH RANDOM AUGMENTATIONS

	SYSU	LEVIR	EGYBCD	GVLM	CLCD	OSCD	*Avg*
None	82.0	**91.0**	**83.5**	88.8	**74.6**	**47.6**	**77.9**
Multistep	81.9	90.8	82.8	**88.9**	73.5	43.6	76.9
Cosine	81.9	90.4	82.9	88.5	73.5	45.3	77.1
Exp.	**82.2**	89.4	81.9	88.2	72.7	46.0	76.7
Linear	81.9	90.3	82.8	88.6	73.2	44.6	76.9
Poly.	81.8	90.4	82.8	88.7	73.3	44.3	76.9

TABLE VI

COMPARISON OF DIFFERENT LOSS FUNCTIONS ON CHANGE DETECTION PERFORMANCE

	SYSU	LEVIR	EGYBCD	GVLM	CLCD	OSCD	*Avg*
CE	**82.0**	91.0	83.5	88.8	74.6	47.6	77.9
Focal	81.8	90.9	83.1	88.9	**75.2**	44.1	77.3
Dice	81.7	91.0	83.3	**89.0**	74.7	**53.0**	**78.8**
Focal+Dice	81.5	90.9	**83.8**	**89.0**	74.4	51.0	78.4
CE+Dice	81.8	**91.1**	83.7	**89.0**	74.1	48.0	77.9

loss [65], have been introduced in recent methods. These are often combined with CE, so we evaluate both individual and combined variants and present the results in Table VI.

While performance varies slightly across datasets, *Dice* loss achieves the best overall result, particularly on the low-resolution OSCD dataset. Dice loss is well-suited for unbalanced data [66] and is usually used as an auxiliary function in related work. However, our results suggest that it is very effective as a standalone primary loss function.

VI. PUTTING IT ALL TOGETHER

This section combines all the findings from our analysis to create a single optimized method called BTC. We present the results when gradually adding improvements to the method in Table VII, and then show that our insights also apply to related methods. Finally, we compare BTC with state-of-the-art approaches.

To ensure practical comparisons while avoiding infeasible amounts of experiments, we incrementally refine the model by applying some of the strongest design choices from each component and report the impact of the best option. A visual summary of the considered elements is presented in Fig. 4.

We begin with a randomly initialized Swin-T trained with CE loss, no augmentations, and no schedulers. The most significant improvement comes from pretraining: introducing *ImageNet1k* pretrained weights boosts performance by 6.4 p.p. Adding *flip* augmentations provides an additional 1.7 p.p. gain. Switching from an ImageNet1k to a *Cityscapes* semantic segmentation pretrained backbone contributes another 0.8 p.p. We also evaluate the *cosine* scheduler since it is the best non-default option from the category. Although it shows no benefits in isolation, it contributes 0.2 p.p when combined with augmentations and different pretraining. Finally, increasing the backbone size to *Swin-B* and replacing CE loss with *Dice* loss yields gains of 0.1 p.p. and 0.2 p.p., respectively.

TABLE VII

RESULTS OF COMBINING OUR INSIGHTS TO REACH THE FINAL OPTIMIZED CHANGE DETECTION ARCHITECTURE. EACH ROW PRESENTS THE $F1$ SCORE AND THE DIFFERENCE COMPARED WITH THE PREVIOUS ROW. STANDARD DEVIATION IS PRESENTED IN SUPPLEMENTARY MATERIAL S4. "+" INDICATES INTRODUCTION OF NEW DESIGN CHOICES, WHILE ⇄ REPRESENTS THE EXCHANGE OF ONE DESIGN CHOICE WITH ANOTHER

	SYSU	LEVIR	EGYBCD	GVLM	CLCD	OSCD	*Avg*
Swin-T	77.0	88.2	76.7	87.5	62.7	37.0	71.5
+ IN1k pre-train	82.0 +5.0	91.0 +2.8	83.5 +6.8	88.8 +1.3	74.6 +11.9	47.6 +10.6	77.9 +6.4
+ Flip augment	81.3 -0.7	91.4 +0.4	85.4 +1.9	90.0 +1.2	78.7 +4.1	51.0 +3.4	79.6 +1.7
⇄ CityS. pre-train	81.0 -0.3	91.5 +0.1	85.9 +0.5	90.1 +0.1	79.1 +0.4	54.8 +3.8	80.4 +0.8
+ Cosine scheduler	81.0 0	91.6 +0.1	**86.2** +0.3	90.5 +0.4	79.4 +0.3	**54.9** +0.1	80.6 +0.2
⇄ Swin-B	81.8 +0.8	**91.7** +0.1	86.0 -0.2	**90.7** +0.2	**81.6** +2.2	52.4 -2.5	80.7 +0.1
⇄ Dice loss	**82.4** +0.6	91.5 -0.2	85.6 -0.4	90.7 0	80.9 -0.7	54.3 +1.9	**80.9** +0.2

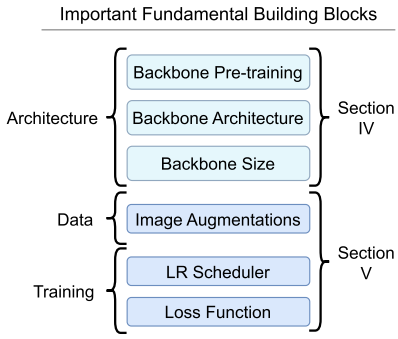


Fig. 4. Summary of important fundamental building blocks analyzed in our study. We consider various architectural, data, and training aspects. The right side of the figure indicates in which section we analyse a specific element: architectural aspects in Section IV, and data with training aspects in Section V.

Together, these optimisations raise the average performance by 9.4 p.p., from 71.5% to 80.9%, with total gains up to 18.2 and 17.3 p.p. on CLCD and OSCD, respectively. The results highlight the impact of basic design choices that can be easily overlooked or inconsistently applied. The substantial improvements from pretraining type selection and simple augmentation show that models can underperform due to sub-optimal defaults rather than inherent architectural limitations. This underscores the need for rigorous baseline evaluations to ensure future change detection research builds on meaningful advancements.

We further analyse our approach through qualitative results in Fig. 5. While performance usually improves with additions, some intermediate steps cause temporary deterioration. Compared with the initial model, the final version better discriminates between irrelevant changes and changes of interest, reducing false positives red regions). It also improves differentiation of visually or semantically similar changes, leading to fewer false negatives blue regions).

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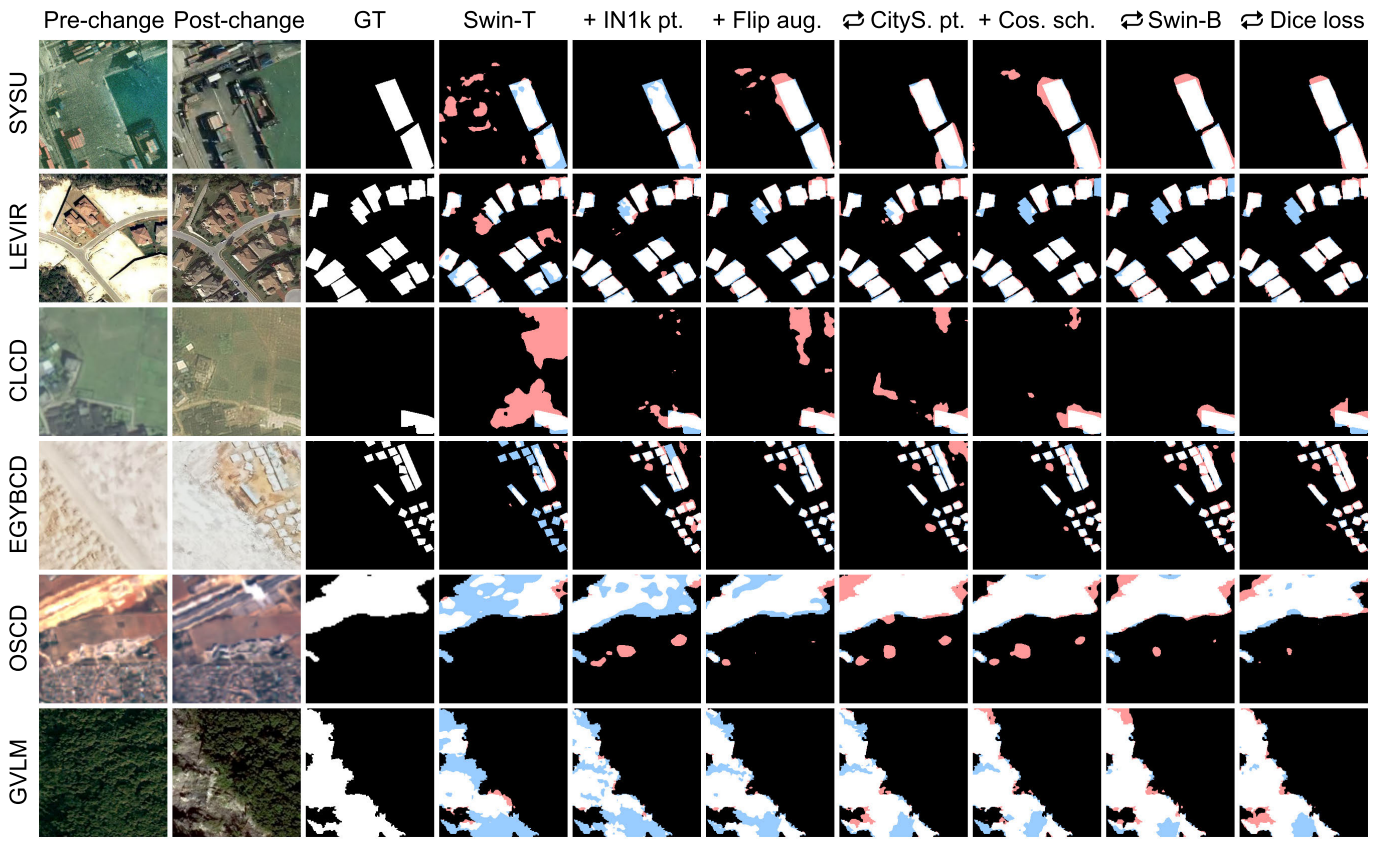


Fig. 5. Qualitative comparison of the predictions made by models with different combined configurations. The pair of considered images is shown in the first and second columns, followed by the ground-truth mask in the third column. The next seven columns depict predictions. False positives are marked in red and false negatives in blue. "+" indicates the introduction of a new design choice, while ⇄ represents the exchange of an existing design choice. Additional qualitative results are given in Supplementary Material S5.

Based on the performed analysis, the quantitative results of combined insights from Table VII, and qualitative results from Fig. 5, it can be concluded that each fundamental element plays an important role in achieving the final performance. Among the most influential are the selection of *backbone architecture and pretraining dataset*, and the *data augmentations* used. When designing change detection models, selecting a strong backbone like Swin, with high spatial resolution and simultaneous global perception, is important. This backbone also greatly benefits from suitable pretraining similar to the change detection task. Due to the imbalance problem of change detection, augmentations that expand the dataset, like flipping and rotating, also creatively contribute to the final result. Finally, it is also important to consider other training elements, among which the loss function and learning rate scheduler play an important role.

A. Transfer to Related Methods

We applied the design choices to related methods to showcase their broader applicability. The results are presented in Table VIII. Each remote sensing foundation model (SeCo, CaCo, SatMAE, GFM, GaSSL, and MTP) is integrated as the backbone within our framework (see Section III). The performance is first assessed under the base configuration: no augmentations, no scheduler, CE loss for foundation models, and the author's default configuration for change-detection-specific models. Additional architecture and hyperparameter details are in Supplementary Material S3-F. We then use flip augmentations, Dice loss, and a cosine scheduler. For change-detection-specific models (FC-Siam-Diff [24] and SwinSUNet [15]), we also exchange their default encoder with Cityscapes pretrained Swin-B used by our approach. This leads to consistent performance gains of several p.p. (full results are in Supplementary Material S3-G).

TABLE VIII

AVERAGE $F1$ ACROSS SIX DATASETS FOR DIFFERENT SOTA METHODS WITH AND WITHOUT USING OPTIMISATIONS FROM OUR INSIGHTS (FULL RESULTS IN SUPPLEMENTARY MATERIAL S3-G AND STANDARD DEVIATION IN SUPPLEMENTARY MATERIAL S4-A)

	SeCo [35]	CaCo [36]	SatMAE [37]	GFM [39]	GeSSL [34]	MTP [40]	FCSiamD [24]	SwinSUN [15]
Base	74.0	75.7	77.1	77.7	75.9	77.2	60.5	78.0
+ Opt.	77.0 +3.0	79.3 +3.6	79.5 +2.4	79.6 +1.9	79.7 +3.8	80.3 +3.1	77.4 +16.9	79.6 +1.6

The large improvement of 16.9 p.p for FC-Siam-Diff and noticeable increase of 1.6 p.p. for SwinSUNet demonstrate that fundamental design choices—such as proper pretraining and training strategies—were previously underutilized in change detection architectures. Despite their architectural innovations, these methods did not fully optimize base components, likely due to the absence of systematic analyses. This highlights the

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value of our findings in guiding future research toward more effective and well-founded model design.

B. Comparison With State of the-Art

To showcase the potency of our findings, we compared BTC to the current state-of-the-art. We include the following remote sensing foundation models: SatMAE [37], GeSSL [34], CaCo [36], SeCo [35], MTP [40], and GFM [39]. We also include methods *specifically designed for change detection*: BIT [1], FC-Siam-Diff [24], ChangeFormer [8], SwinSUNet [15], and BiFA [6]. Since remote sensing foundation models don't have a specific training protocol, we utilize our optimized framework. Change detection-specific architectures already define the training protocol, so we use their original setup.

In accordance with our protocol, we rerun all experiments with three different random seeds and report the average result. We also report standard deviations for dataset averages here, while complete deviation results are in Supplementary Material S4-B and implementation details are in Supplementary Material S3-F.

1) Change Detection Performance: Table IX presents a comparison of our approach to state-of-the-art methods. By leveraging all insights from our analysis, BTC surpasses previously proposed models on the CLCD, GVLM, and OSCD datasets. It also achieves competitive results on SYSU, LEVIR, and EGYBCD and achieves the best average performance across datasets.

BTC T outperforms SwinSUNet [15] by 2.5 p.p. using the same Swin-T backbone, a simpler fusion module and decoder, but a more optimal training setup. This reiterates our claim that previous methods have not sufficiently explored the base design choices. The gap is reduced to 1.3 p.p. if we apply our optimisations to SwinSUNet (as presented in Section VI-A). This means BTC still performs better, suggesting that some complex fusion and decoder blocks might even harm the performance when paired with optimal training setups.

Remote sensing pretrained models generally outperform change detection methods on smaller datasets like CLCD and OSCD, where limited data hinders the training of architecturally advanced fusion and decoder modules. Yet, our approach outperforms even foundation models despite using weights pretrained on natural images. As discussed in Section IV-A and in [48], this highlights the importance of task selection in pretraining, not just representation quality. Segmentation pretrained weights benefit change detection, which may also explain the strong performance of MTP [40], where segmentation is one of the pretraining tasks.

2) Computational Efficiency: While our approach and foundation models have larger parameter sizes, their inference speed (measured in frames per second—FPS—fourth column in Table IX) can still match smaller architectures on efficient modern hardware. Details on the computational efficiency evaluation protocol with additional metrics are in Supplementary Material S3-H.

3) Qualitative Results: Qualitative results in Fig. 6 show predictions for our method compared to the best-performing remote sensing foundation model MTP [40] and the best-performing change detection architecture SwinSUNet [15]. While all methods predict visually similar results, our method, in some cases, produces fewer false positive regions.

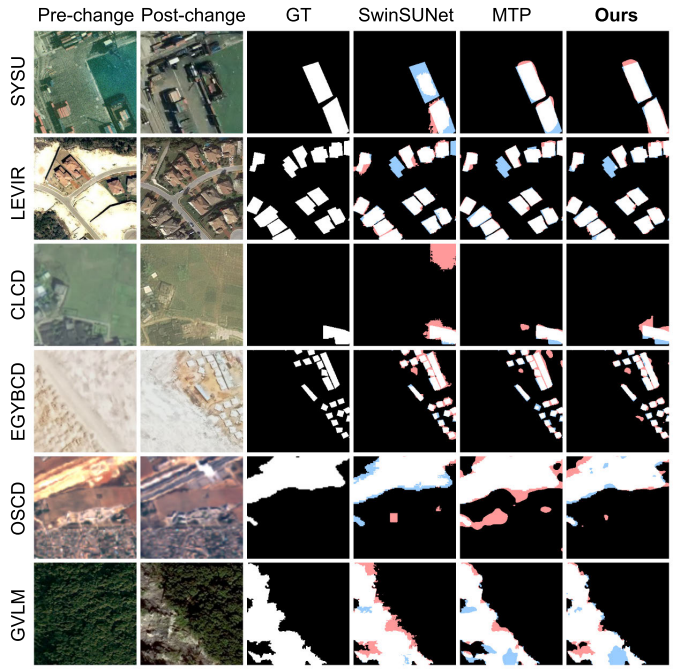


Fig. 6. Qualitative comparison of the predictions made by our method and best-performing related methods: SwinSUNet (change-detection-specific) and MTP (foundation model). The pair of considered images is shown in the first and second columns, followed by the ground truth mask and predictions for each method. False positives are marked in red and false negatives in blue. Qualitative results for all other models are given in Supplementary Material S5-B.

C. Discussion

The results emphasize the critical role of carefully selecting fundamental elements of change detection methods. While the optimisations we highlight may appear standard, their application remains inconsistent, likely due to their perceived simplicity. Improved basic and standard elements potentially overshadow other architectural additions in new methods because their impact was not well established. This also raises the question of how much further the performance could be improved with advancements built with better fundamental components.

BTC and the accompanying analysis provide a strong and transparent baseline for both change detection architectures and remote sensing pretraining. Evaluation of future methods could benefit from clarity, preventing situations where novel architectural additions potentially reduce the performance compared with the baseline, utilizing the same backbone but simpler fusion and decoder architecture. The results also strengthen the need for pretraining strategies tailored to the task—potentially leveraging segmentation-inspired techniques—while integrating remote sensing-specific properties to outperform natural-image segmentation pretraining.

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TABLE IX

COMPARISON OF THE PROPOSED METHOD TO THE STATE-OF-THE-ART ON SIX DIFFERENT DATASETS. WE REPORT THE MEAN $F1$ OF THREE RUNS WITH DIFFERENT RANDOM SEEDS FOR EACH DATASET AND AN AVERAGE WITH STANDARD DEVIATION ACROSS ALL DATASETS (STANDARD DEVIATION FOR INDIVIDUAL DATASETS IS IN SUPPLEMENTARY MATERIAL S4-B). FIRST, SECOND, AND THIRD PLACE RESULTS ARE MARKED. REMOTE SENSING PRETRAINING BASED METHODS ARE HIGHLIGHTED IN BLUE, WHILE CHANGE DETECTION SPECIFIC ARCHITECTURES ARE MARKED WITH ORANGE. FPS WAS BENCHMARKED ON AN NVIDIA A100-SXM4 40GB GPU (DETAILS IN SUPPLEMENTARY MATERIAL S3-H)

	Backbone	Pre-train	FPS [img/s]	Param. [M]	SYSU	LEVIR	EGYBCD	GVLM	CLCD	OSCD	*Avg & std*
FCS-Diff [24]	UNet	-	170.1	1.4	70.8	81.8	42.3	74.3	54.1	39.4	60.5 ±0.5
BIT [1]	RN-18	IN1k	57.7	12.4	77.0	90.0	75.9	88.8	63.4	42.6	73.0 ±0.4
ChangeFormer [8]	MiT-B0	-	36.2	41.0	77.9	89.5	77.9	88.5	60.8	48.1	73.8 ±0.3
BiFA [6]	MiT-B0	ADE20k	32.2	9.9	**83.8**	89.5	83.5	89.0	74.5	37.4	76.3 ±0.8
SeCo [35]	RN-50	RS	100.7	64.0	78.9	90.6	84.1	90.4	68.7	49.1	77.0 ±0.2
SwinSUNet [15]	Swin-T	IN1k	33.1	43.6	76.6	89.3	83.7	90.0	75.8	52.8	78.0 ±0.4
CaCo [36]	RN-50	RS	100.9	64.0	80.2	90.9	84.8	90.3	77.9	51.9	79.3 ±0.0
SatMAE [37]	ViT-L	RS	71.2	322.6	81.4	91.4	**85.9**	89.8	79.2	49.5	79.5 ±0.2
GFM [39]	Swin-B	RS	44.9	120.5	81.2	89.8	85.0	89.8	77.5	54.1	79.6 ±0.3
GeSSL [34]	RN-50	RS	100.4	64.0	80.2	91.2	85.4	90.2	78.1	53.1	79.7 ±0.2
MTP [40]	ViT-B + RVSA	RS	31.2	107.8	81.3	**91.7**	**85.9**	89.9	80.3	52.8	80.3 ±0.1
BTC T (Ours)	Swin-T	CityS.	57.8	58.9	81.6	91.4	85.6	90.6	79.5	**54.3**	80.5 ±0.2
BTC B (Ours)	Swin-B	CityS.	32.4	120.1	82.4	91.5	85.6	**90.7**	**80.9**	**54.3**	**80.9** ±0.0

While our final optimized model employs a Swin architecture, we recognize that alternative architectures might exhibit different optimal configurations concerning various design options. Nevertheless, our study emphasizes the substantial influence of configurations such as pretraining and augmentations (see Table VII). We anticipate that future research will take these aspects into account more diligently, as they were not sufficiently explored before.

VII. CONCLUSION

In this work, we present BTC, a simple yet strong model for change detection, which achieves state-of-the-art performance across six datasets. Our approach highlights the importance of fundamental design choices that drive performance, supported by an extensive empirical analysis. By analyzing backbone configurations and training strategies, we offer valuable insights and guidelines for future methods. The impact of optimisations resulting from our analysis is evident in the base model's performance improvement of F1 from 71.5% to 80.9% across six diverse change detection datasets.

We further demonstrate the impact of our observations by applying them to related models, significantly improving their performance. The results suggest that previous approaches have often overlooked the contribution of these fundamental choices, potentially due to a lack of a detailed analysis. We thus believe that our findings and the BTC model lay a strong foundation for future advancements in the field.

A key direction for future work is extending these insights to multimodal data and other remote sensing tasks. This includes adapting pretrained weights for SAR, multispectral, and hyperspectral applications and refining pretraining strategies to better align with change detection and other dense prediction tasks, such as segmentation.

REFERENCES

[1] H. Chen, Z. Qi, and Z. Shi, "Remote sensing image change detection with transformers," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, pp. 1–14, 2021.

[2] R. C. Daudt, B. Le Saux, A. Boulch, and Y. Gousseau, "Urban change detection for multispectral Earth observation using convolutional neural networks," in *Proc. IEEE Int. Geosci. Remote Sens. Symp.*, Jul. 2018, pp. 2115–2118.

[3] Z. Zheng, Y. Zhong, L. Zhang, and S. Ermon, "Segment any change," in *Proc. Adv. Neural Inf. Process. Syst.*, Feb. 2024, pp. 81204–81224.

[4] Z. Zhu, S. Qiu, and S. Ye, "Remote sensing of land change: A multifaceted perspective," *Remote Sens. Environ.*, vol. 282, Dec. 2022, Art. no. 113266.

[5] R. Hänsch and M. A. Chaurasia, "Earth observation and machine learning for climate change," in *Proc. IGARSS - IEEE Int. Geosci. Remote Sens. Symp.*, Jul. 2024, pp. 1676–1682.

[6] H. Zhang et al., "BiFA: Remote sensing image change detection with bitemporal feature alignment," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5614317.

[7] W. G. Chaminda Bandara, N. G. Nair, and V. M. Patel, "DDPM-CD: Denoising diffusion probabilistic models as feature extractors for remote sensing change detection," in *Proc. IEEE/CVF Winter Conf. Appl. Comput. Vis. (WACV)*, Feb. 2025, pp. 5250–5262.

[8] W. G. C. Bandara and V. M. Patel, "A transformer-based Siamese network for change detection," in *Proc. IEEE Int. Geosci. Remote Sens. Symp.*, Jul. 2022, pp. 207–210.

[9] Y. Feng, J. Jiang, H. Xu, and J. Zheng, "Change detection on remote sensing images using dual-branch multilevel intertemporal network," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 4401015.

[10] Z. Li et al., "Lightweight remote sensing change detection with progressive feature aggregation and supervised attention," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5602812.

[11] Z. Liu et al., "Swin transformer: Hierarchical vision transformer using shifted windows," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2021, pp. 10012–10022.

[12] Z. Liu et al., "Swin transformer V2: Scaling up capacity and resolution," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2022, pp. 12009–12019.

[13] Z. Liu, H. Mao, C.-Y. Wu, C. Feichtenhofer, T. Darrell, and S. Xie, "A ConvNet for the 2020s," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jun. 2022, pp. 11976–11986.

[14] Z. Zheng, A. Ma, L. Zhang, and Y. Zhong, "Change is everywhere: Single-temporal supervised object change detection in remote sensing imagery," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2021, pp. 15193–15202.

[15] C. Zhang, L. Wang, S. Cheng, and Y. Li, "SwinSUNet: Pure transformer network for remote sensing image change detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5224713.

[16] H. Chen, J. Song, C. Han, J. Xia, and N. Yokoya, "ChangeMamba: Remote sensing change detection with spatiotemporal state space model," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 4409720.

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[17] L. Heckler, R. König, and P. Bergmann, "Exploring the importance of pretrained feature extractors for unsupervised anomaly detection and localization," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. Workshops (CVPRW)*, Jun. 2023, pp. 2917–2926.

[18] M. Oršic, I. Krešo, P. Bevandic, and S. Šegvic, "In defense of pre-trained ImageNet architectures for real-time semantic segmentation of road-driving images," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2019, pp. 12599–12608.

[19] H. Zhang et al., "DINO: DETR with improved DeNoising anchor boxes for end-to-end object detection," in *Proc. Int. Conf. Learn. Represent.*, Jan. 2022, pp. 8587–8605.

[20] B. Rolih, M. Fuč ka, and D. Skoč aj, "SuperSimpleNet: Unifying unsupervised and supervised learning for fast and reliable surface defect detection," in *Proc. Int. Conf. Pattern Recognit.*, Dec. 2024, pp. 47–65.

[21] A. Singh, "Review article digital change detection techniques using remotely-sensed data," *Int. J. Remote Sens.*, vol. 10, no. 6, pp. 989–1003, Jun. 1989.

[22] B. Le Saux and H. Randrianarivo, "Urban change detection in SAR images by interactive learning," in *Proc. IEEE Int. Geosci. Remote Sens. Symp.*, Jul. 2013, pp. 3990–3993.

[23] N. Metzger, M. Ö. Türkoglu, R. C. Daudt, J. D. Wegner, and K. Schindler, "Urban change forecasting from satellite images," *PFG-J. Photogramm., Remote Sens. Geoinf. Sci.*, vol. 91, no. 6, pp. 443–452, Dec. 2023.

[24] R. C. Daudt, B. Le Saux, and A. Boulch, "Fully convolutional Siamese networks for change detection," in *Proc. 25th IEEE Int. Conf. Image Process. (ICIP)*, Oct. 2018, pp. 4063–4067.

[25] A. Dosovitskiy et al., "An image is worth 16×16 words: Transformers for image recognition at scale," in *Proc. Int. Conf. Learn. Represent.*, Jan. 2020, pp. 611–631.

[26] A. Gu and T. Dao, "Mamba: Linear-time sequence modeling with selective state spaces," in *Proc. 1st Conf. Lang. Modeling*, Jan. 2023, pp. 1–32. [Online]. Available: https://openreview.net/forum?id=tEYskw1VY2

[27] Y. Liu et al., "VMamba: Visual state space model," in *Proc. Adv. Neural Inf. Process. Syst.*, Jan. 2024, pp. 103031–103063. [Online]. Available: https://openreview.net/forum?id=ZgtLQQR1K7

[28] H. Chen and Z. Shi, "A spatial–temporal attention-based method and a new dataset for remote sensing image change detection," *Remote Sens.*, vol. 12, no. 10, p. 1662, 2020.

[29] M. Noman et al., "Remote sensing change detection with transformers trained from scratch," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 4704214.

[30] J. Deng, W. Dong, R. Socher, L.-J. Li, K. Li, and L. Fei-Fei, "ImageNet: A large-scale hierarchical image database," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit.*, Jun. 2009, pp. 248–255.

[31] K. Li, X. Cao, and D. Meng, "A new learning paradigm for foundation model-based remote-sensing change detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5610112.

[32] G. Astruc, N. Gonthier, C. Mallet, and L. Landrieu, "AnySat: One Earth observation model for many resolutions, scales, and modalities," 2024, *arXiv:2412.14123*.

[33] G. Astruc, N. Gonthier, C. Mallet, and L. Landrieu, "OmniSat: Self-supervised modality fusion for Earth observation," in *Proc. Eur. Conf. Comput. Vis.*, Oct. 2024, pp. 409–427.

[34] K. Ayush et al., "Geography-aware self-supervised learning," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2021, pp. 10181–10190.

[35] O. Manas, A. Lacoste, X. Giro-i-Nieto, D. Vazquez, and P. Rodriguez, "Seasonal contrast: Unsupervised pre-training from uncurated remote sensing data," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2021, pp. 9414–9423.

[36] U. Mall, B. Hariharan, and K. Bala, "Change-aware sampling and contrastive learning for satellite images," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2023, pp. 5261–5270.

[37] Y. Cong et al., "SatMAE: Pre-training transformers for temporal and multi-spectral satellite imagery," in *Proc. Adv. Neural Inf. Process. Syst.*, Jan. 2022, pp. 197–211.

[38] M. Tang, A. Cozma, K. Georgiou, and H. Qi, "Cross-scale mae: A tale of multiscale exploitation in remote sensing," in *Proc. Adv. Neural Inf. Process. Syst.*, vol. 36, 2024, pp. 20054–20066.

[39] M. Mendieta, B. Han, X. Shi, Y. Zhu, and C. Chen, "Towards geospatial foundation models via continual pretraining," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2023, pp. 16806–16816.

[40] D. Wang et al., "MTP: Advancing remote sensing foundation model via multitask pretraining," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 17, pp. 11632–11654, 2024.

[41] D. Wang, J. Zhang, B. Du, G.-S. Xia, and D. Tao, "An empirical study of remote sensing pretraining," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5608020.

[42] Y. Zhang, Y. Zhao, Y. Dong, and B. Du, "Self-supervised pretraining via multimodality images with transformer for change detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5402711.

[43] M. Leenstra, D. Marcos, F. Bovolo, and D. Tuia, "Self-supervised pre-training enhances change detection in Sentinel-2 imagery," in *Proc. Int. Conf. Pattern Recognit.*, vol. 12667. Cham, Switzerland: Springer, 2021, pp. 578–590.

[44] Y. Quan et al., "Unified building change detection pre-training method with masked semantic annotations," *Int. J. Appl. Earth Observ. Geoinf.*, vol. 120, Jun. 2023, Art. no. 103346.

[45] Z. Zheng, S. Tian, A. Ma, L. Zhang, and Y. Zhong, "Scalable multi-temporal remote sensing change data generation via simulating stochastic change process," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2023, pp. 21818–21827.

[46] T. He, Z. Zhang, H. Zhang, Z. Zhang, J. Xie, and M. Li, "Bag of tricks for image classification with convolutional neural networks," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2019, pp. 558–567.

[47] W. Yu et al., "Bag of tricks for training data extraction from language models," in *Proc. Int. Conf. Mach. Learn.*, Jan. 2023, pp. 40306–40320.

[48] J. Sosa et al., "How effective is pre-training of large masked autoencoders for downstream earth observation tasks?" in *Proc. Workshop Mach. Vis. Earth Observ. Environ. Monitor.*, 2024, pp. 1–12.

[49] I. Corley, C. Robinson, and A. Ortiz, "A change detection reality check," 2024, *arXiv:2402.06994*.

[50] K. He, X. Zhang, S. Ren, and J. Sun, "Deep residual learning for image recognition," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2016, pp. 770–778.

[51] T. Xiao, Y. Liu, B. Zhou, Y. Jiang, and J. Sun, "Unified perceptual parsing for scene understanding," in *Proc. Eur. Conf. Comput. Vis. (ECCV)*, 2018, pp. 418–434.

[52] Q. Shi, M. Liu, S. Li, X. Liu, F. Wang, and L. Zhang, "A deeply supervised attention metric-based network and an open aerial image dataset for remote sensing change detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5604816.

[53] S. Holail, T. Saleh, X. Xiao, and D. Li, "AFDE-Net: Building change detection using attention-based feature differential enhancement for satellite imagery," *IEEE Geosci. Remote Sens. Lett.*, vol. 20, pp. 1–5, 2023.

[54] X. Zhang, W. Yu, M.-O. Pun, and W. Shi, "Cross-domain landslide mapping from large-scale remote sensing images using prototype-guided domain-aware progressive representation learning," *ISPRS J. Photogramm. Remote Sens.*, vol. 197, pp. 1–17, Mar. 2023.

[55] M. Liu, Z. Chai, H. Deng, and R. Liu, "A CNN-transformer network with multiscale context aggregation for fine-grained cropland change detection," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 15, pp. 4297–4306, 2022.

[56] I. Loshchilov and F. Hutter, "Decoupled weight decay regularization," in *Proc. Int. Conf. Learn. Represent.*, Jan. 2017, pp. 4061–4078.

[57] B. Zhou, H. Zhao, X. Puig, S. Fidler, A. Barriuso, and A. Torralba, "Scene parsing through ADE20K dataset," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jul. 2017, pp. 633–641.

[58] M. Cordts et al., "The cityscapes dataset for semantic urban scene understanding," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2016, pp. 3213–3223.

[59] T. Lin et al., "Microsoft COCO: Common objects in context," in *Proc. Eur. Conf. Comput. Vis.*, 2014, pp. 740–755.

[60] P. Helber, B. Bischke, A. Dengel, and D. Borth, "EuroSAT: A novel dataset and deep learning benchmark for land use and land cover classification," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 12, no. 7, pp. 2217–2226, Jul. 2019.

[61] P. Helber, B. Bischke, A. Dengel, and D. Borth, "Introducing eurosat: A novel dataset and deep learning benchmark for land use and land cover classification," in *Proc. IEEE Int. Geosci. Remote Sens. Symp.*, Jul. 2018, pp. 204–207.

[62] E. Rolf, K. Klemmer, C. Robinson, and H. Kerner, "Position: Mission critical–satellite data is a distinct modality in machine learning," in *Proc. Int. Conf. Mach. Learn.*, 2024, pp. 42691–42706. [Online]. Available: https://openreview.net/forum?id=PQ0ERKKYJu

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[63] I. Loshchilov and F. Hutter, "SGDR: Stochastic gradient descent with warm restarts," in *Proc. Int. Conf. Learn. Represent.*, Jan. 2016, pp. 1769–1784. [Online]. Available: https://openreview.net/forum?id=Skq89Scxx

[64] F. Milletari, N. Navab, and S. Ahmadi, "V-Net: Fully convolutional neural networks for volumetric medical image segmentation," in *Proc. 4th Int. Conf. 3D Vis. (3DV)*, Oct. 2016, pp. 565–571.

[65] T.-Y. Lin, P. Goyal, R. Girshick, K. He, and P. Dollár, "Focal loss for dense object detection," in *Proc. IEEE Int. Conf. Comput. Vis. (ICCV)*, Oct. 2017, pp. 2980–2988.

[66] C. H. Sudre, W. Li, T. Vercauteren, S. Ourselin, and M. J. Cardoso, "Generalised dice overlap as a deep learning loss function for highly unbalanced segmentations," in *Proc. Int. Workshop Deep Learn. Med. Image Anal.* Cham, Switzerland: Springer, 2017, pp. 240–248.



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